

A Search for Anomalous Spectral Energy Distributions in High-Redshift Galaxies with JWST: A Data-Driven Approach

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ABSTRACT

The James Webb Space Telescope (JWST) has produced the deepest and most complete rest-frame optical/near-infrared photometric catalogs of high-redshift galaxies to date, enabling systematic searches for sources whose spectral energy distributions (SEDs) deviate from standard population-synthesis expectations. We present a data-driven pipeline that identifies statistically anomalous SEDs among $z \gtrsim 0.5$ galaxies in the CEERS and JADES JWST/NIRCam photometric catalogs. Photometric redshifts and best-fit template SEDs are derived with EAZY; per-band residuals between observed and best-fit photometry are then scored for anomalies using an ensemble of Isolation Forest and UMAP+DBSCAN, and cross-matched against known active galactic nucleus (AGN) and emission-line-galaxy catalogs to separate genuine outliers from well-understood astrophysical contaminants. We report the resulting anomaly rate as a function of redshift and characterize the residual “unexplained” population that survives contaminant removal. We interpret this population within a deliberately conservative framework for galactic-scale technosignature and biosignature search limits, noting that even a well-characterized null result constitutes a quantitative upper limit on non-standard astrophysical processes at cosmological scale. This placeholder abstract will be updated once the Phase 3/4 analysis on the full real MAST catalogs is complete.

Keywords: galaxies: high-redshift — galaxies: photometry — methods: data analysis — methods: statistical — astrobiology

1. INTRODUCTION

[PLACEHOLDER — JWST context] The James Webb Space Telescope has transformed the study of galaxy evolution at $z > 5$ by providing sensitive, high-angular-resolution NIRCam and MIRI imaging across a wavelength baseline unattainable from the ground or with prior space missions. Large-area imaging surveys such as CEERS (Finkelstein & et al. 2023), JADES (Eisenstein & et al. 2023), and COSMOS-Web (Casey & et al. 2023) have assembled photometric catalogs of hundreds of thousands of galaxies spanning cosmic noon through the epoch of reionization. These catalogs are large and homogeneous enough to support systematic, population-level searches for statistically rare objects, rather than only targeted follow-up of individually identified curiosities.

[PLACEHOLDER — ML anomaly detection in astrophysics] Unsupervised machine learning methods have an established track record of surfacing unusual sources in large astronomical datasets that would be impractical to find by visual inspection alone, from outlier detection in SDSS spectra (Baron & Poznanski 2017) to searches for spectroscopically unusual quasars (Meusinger & et al. 2012). Applying similar techniques to JWST photometric residuals — the difference between observed fluxes and the best-fit template SED at each source’s photometric redshift — offers a principled, reproducible way to flag galaxies whose broadband colors are not well described by standard stellar population synthesis models (Reyes & et al. 2023), without requiring a predefined list of what an “anomaly” should look like.

[PLACEHOLDER — astrobiology motivation] A small subset of any such anomaly sample may, in principle, be astrophysically unexplained even after accounting for known contaminants such as AGN activity or strong emission-line contamination of broadband photometry. While the overwhelming prior expectation is

that such residual anomalies reflect systematics in photometry, photometric redshifts, or template libraries, we adopt the deliberately conservative position that a well-characterized, reproducible anomaly pipeline is also the correct tool with which to place quantitative, galactic-scale upper limits on exotic interpretations, including technosignatures (Wright et al. 2014) and biosignature-adjacent astrophysical scenarios (Lingam & Loeb 2019; Lacki 2016). This paper does not claim any detection; its purpose is methodological: to demonstrate a reproducible anomaly-detection pipeline and to characterize, at current survey depth, how large an unexplained population remains after standard astrophysical explanations are exhausted.

2. DATA

2.1. CEERS

[PLACEHOLDER] We use the CEERS Data Release 1.0 photometric and physical-parameter catalog (Finkelstein & et al. 2023), comprising NIRC*am* imaging across the CEERS pointings within the Extended Groth Strip. The catalog provides PSF-matched aperture photometry in the F090W, F115W, F150W, F200W, F277W, F356W, F410M, and F444W bands, along with LePHARE photometric redshift estimates. [PLACEHOLDER: final source counts, magnitude limits, and areal coverage to be filled in once the full-catalog run completes.]

2.2. JADES

[PLACEHOLDER] We additionally draw on the JWST Advanced Deep Extragalactic Survey (JADES) Data Release 1 (Eisenstein & et al. 2023), which reaches substantially fainter magnitude limits than CEERS over a smaller area, providing a complementary deep pencil-beam sample. [PLACEHOLDER: JADES retrieval is not yet complete at the time of this scaffold; final sample statistics to follow.]

2.3. Data Reduction

[PLACEHOLDER] Raw catalogs are retrieved directly from their public data release locations, normalized to a common column schema (right ascension, declination, photometric redshift, and per-band flux/flux-uncertainty pairs), and converted to a common flux unit (μJy). Sources are required to fall within the survey’s science redshift window and to have a minimum number of valid photometric bands (Section 3.1) before proceeding to SED fitting.

3. METHODS

3.1. SED Fitting

[PLACEHOLDER] Photometric redshifts and best-fit template SEDs are derived using EAZY (Brammer et al. 2008) with the `tweak_fsp_QSF_12_v3` template set, itself built on Bruzual & Charlot stellar population synthesis models (Bruzual & Charlot 2003). [PLACEHOLDER: describe final adopted redshift grid, template combination weighting, and any priors applied.]

3.2. Residual Extraction

[PLACEHOLDER] For each source, per-band photometric residuals are computed as $(f_{\text{obs}} - f_{\text{model}})/\sigma_{\text{obs}}$ relative to a smooth continuum model fit through the observed photometry, following the general SED-fitting residual framework reviewed by Conroy (2013). These residual vectors form the feature space for the anomaly detectors described below.

3.3. Anomaly Detection

[PLACEHOLDER] We score each source’s residual vector using an ensemble of two complementary unsupervised anomaly detectors: an Isolation Forest, which isolates anomalous points via random recursive partitioning, and a UMAP dimensionality reduction followed by DBSCAN density-based clustering, which flags points that fail to join any density-connected cluster in the reduced embedding. Combining a global, tree-based method with a local, density-based method mirrors approaches used elsewhere in astronomical outlier detection (Baron & Poznanski 2017; Reyes & et al. 2023).

3.4. Contaminant Removal

[PLACEHOLDER] Flagged sources are cross-matched on sky position against a reference AGN catalog (Milliquas) to identify anomalies plausibly explained by AGN continuum/emission-line contamination, and are separately checked for redshifted strong emission lines (e.g., Ly α) falling within a broadband filter, which can locally inflate a single band’s flux and masquerade as an SED anomaly (Meusinger & et al. 2012). Sources matching either criterion are retained in the catalog but flagged as “explained” anomalies, distinct from the residual “unexplained” population discussed in Section 4.3.

4. RESULTS

4.1. Anomaly Rate

[PLACEHOLDER: overall anomaly rate, broken down by survey and by which detector(s) flagged each source, to be filled in once the full pipeline run on real MAST catalogs completes. A summary table and figure will be added here once available.]

4.2. Redshift Evolution

[PLACEHOLDER: anomaly rate as a function of photometric redshift, binned in $\Delta z = 0.5$ shells, with Poisson uncertainties on the flagged count per bin (see [results/figures/anomaly_rate_vs_redshift.pdf](#)).]

4.3. Unexplained Outliers

[PLACEHOLDER: characterization of the subset of flagged sources that match neither the AGN nor emission-line contaminant criteria, including a summary table of the highest-ranked "unexplained" outliers by ensemble anomaly score (see [results/tables/top_outliers.tex](#)).]

5. DISCUSSION

[PLACEHOLDER] We discuss the interpretation of the unexplained anomaly population in the context of (i) residual systematics in photometry, photometric redshifts, or template libraries; (ii) known but uncatalogued astrophysical contaminants (e.g., strong lensing, blended sources, rare transient contamination); and (iii) the deliberately conservative galactic-scale technosignature/biosignature interpretive framework motivated in Section 1 (Wright et al. 2014; Lingam & Loeb 2019; Lacki 2016). We emphasize that a well-characterized null result is itself a meaningful, quantitative constraint on the prevalence of non-standard processes at current survey depth and sample size.

6. CONCLUSIONS

[PLACEHOLDER] We have presented a reproducible, ensemble machine-learning pipeline for identifying anomalous SEDs among high-redshift JWST galaxies, and applied it to the CEERS and (forthcoming) JADES photometric catalogs. [PLACEHOLDER: summarize final anomaly rate, fraction explained by known contaminants, and headline redshift-dependence result once the full analysis is complete. Future work will extend the pipeline to COSMOS-Web and incorporate spectroscopic follow-up of the highest-confidence unexplained outliers.]

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9 CEERS: 1345], #[PLACEHOLDER — JADES program
10 ID], and #[PLACEHOLDER — COSMOS-Web pro-
11 gram ID] via DOI.

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